
A Mission-Oriented Tool for System-of-Systems Modeling

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Outline



- Introduction
- Modeling missions of SoS
- mKAOS language and tool
- An illustrative example
- Concluding remarks

The role of missions in SoS

- Missions: a two-level concern in the SoS context
 - Individual missions accomplished by constituent systems
 - Global missions accomplished by the SoS
- Missions in SoS can be related to
 - Required capabilities of constituent systems
 - Interactions among constituent systems that contribute to the accomplishment of global missions
 - High-level concerns: no details about implementation

The role of missions in SoS

Mission modeling as the **starting activity** for developing SoS

Mission model

Representation of

- (1) global missions** assigned to the SoS
- (2) individual missions** assigned to constituent systems
- (3) required capabilities** for accomplishing missions
- (4) interactions** among constituent systems

The role of missions in SoS

Use of mission models **throughout** the whole SoS development process

- Use of mission models as a basis for elaborating **architectural models**
- Verification of **compliance** of an SoS software architecture with the defined missions

Existing gaps

Lack of specific languages, models, and tools for defining missions in the SoS context

- Existing approaches for SoS do not handle mission modeling
- Solutions for modeling missions of systems do not address important concerns related to SoS



E. Silva, E. Cavalcante, T. Batista, F. Oquendo, F. C. Delicato, and P. F. Pires. **On the characterization of missions of systems-of-systems**. Proceedings of the 2014 European Conference on Software Architecture Workshops. New York, NY, USA: ACM, 2014.

Contributions

- **mKAOS**

graphical modeling language for specifying missions

- **mKAOS Studio**

open-source tool for assisting SoS designers in mission modeling by using the mKAOS language

Our approach

Extension of the KAOS language and methodology for requirements engineering

- KAOS provide elements able to represent a significant set of the concepts identified in the conceptual model
- Adoption of KAOS in both academia and industry
- Separation of concerns
- Overlapping different models to cross-view a system
- Main concern: goals ↔ missions

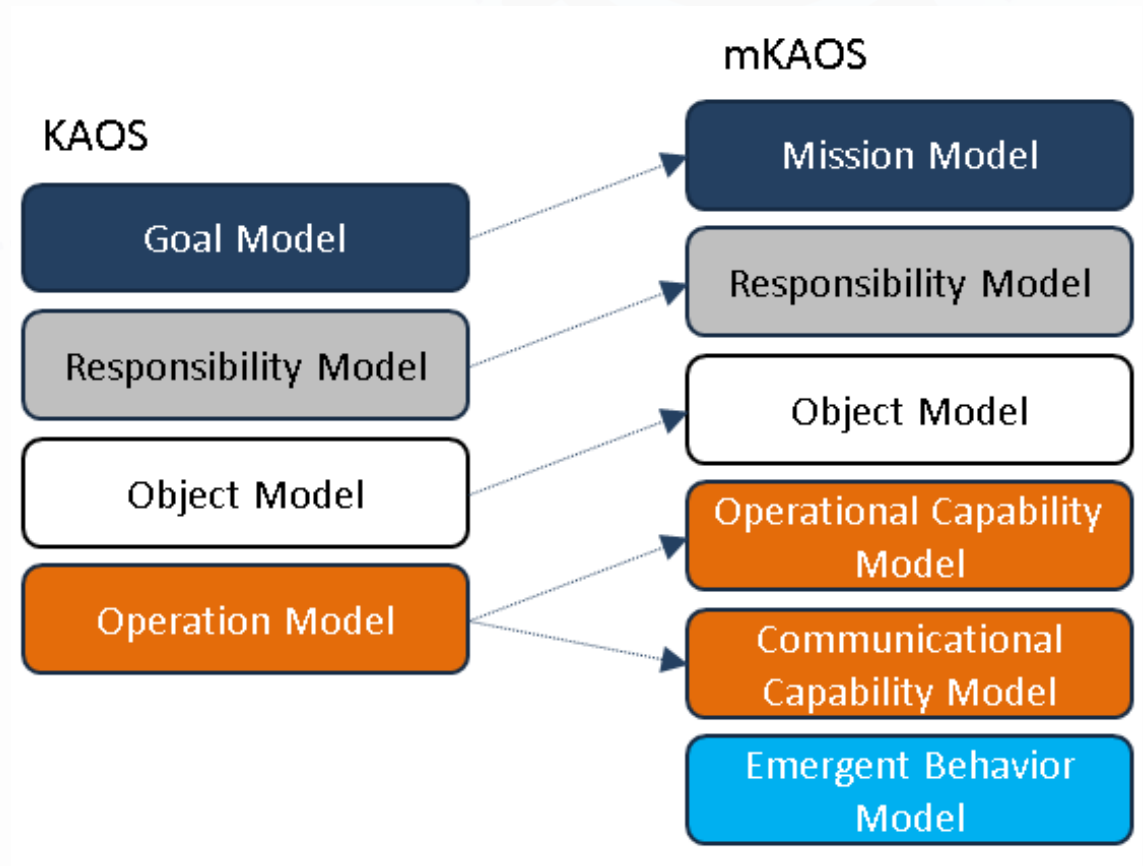
The mKAOS language

Aims

- Description of both **global and individual missions**
- **Assignment of individual missions** to constituent systems
- Correlation of global missions with **emergent behaviors** arisen from interactions among constituent systems
- Description of **capabilities** of constituent systems
- Specification of additional **mission-related information**

The mKAOS language

mKAOS models



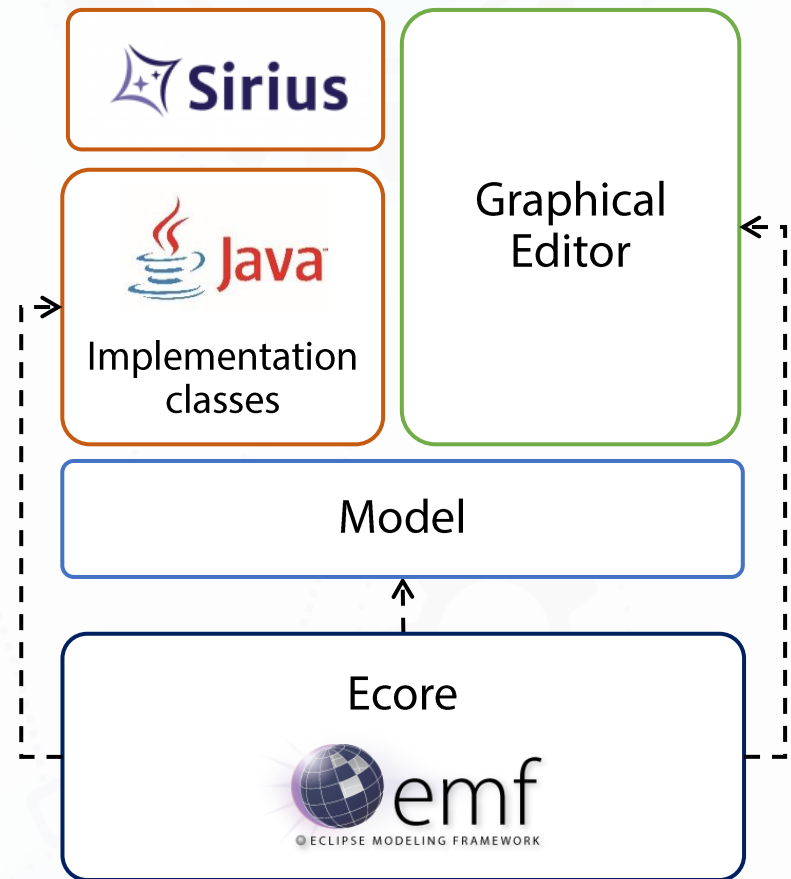
The mKAOS language

mKAOS models

Model	Definition
Mission model	Description of individual missions and global missions
Responsibility Model	Description of both constituent systems and assignment of them to the respective individual missions
Object Model	Description of operational objects (entities and events) used within the scope of the SoS
Operational Capability model	Description of operations performed by constituent systems (operational capabilities)
Communicational Capability Model	Description of interactions among constituent systems (communicational capabilities)
Emergent Behavior Model	Description of [foreseen] emergent behaviors and their relationship with global missions

The mKAOS Studio tool

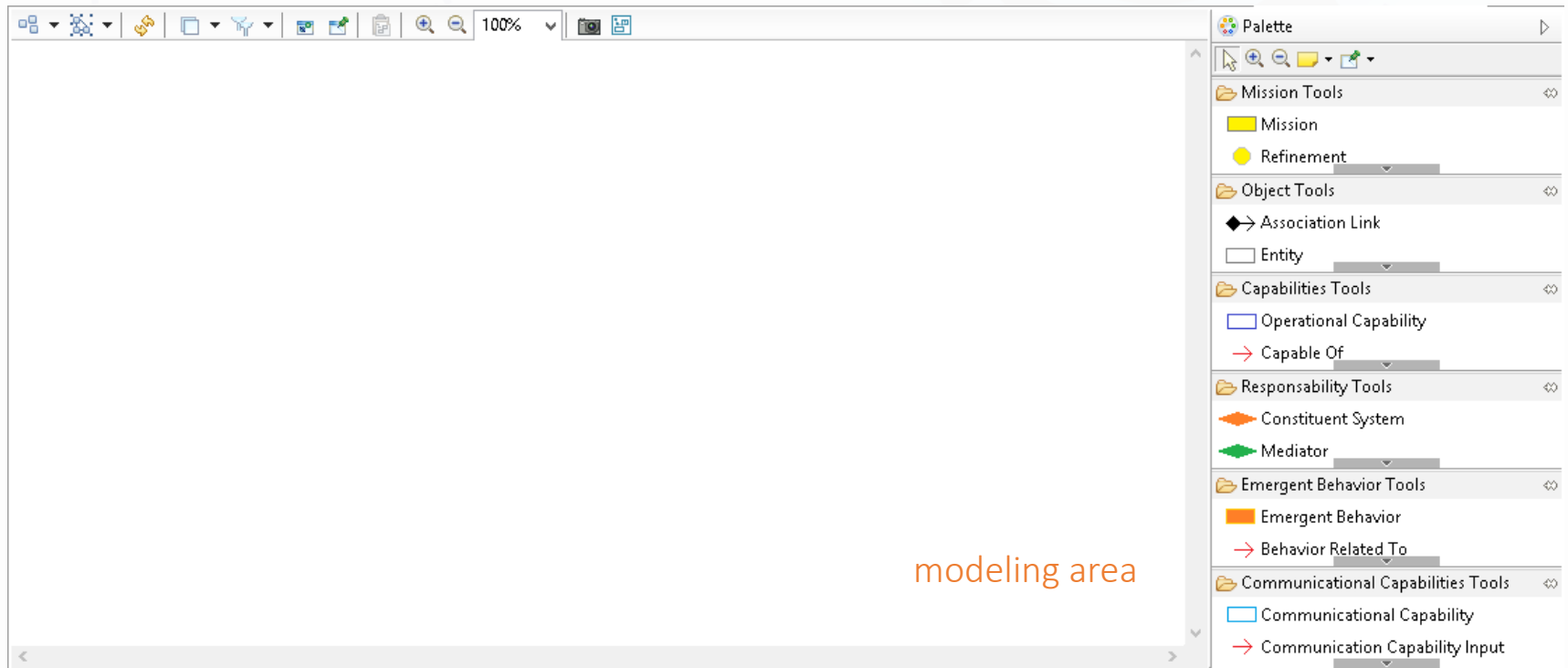
- Eclipse plug-in
 - Eclipse Modeling Framework (EMF)
 - Sirius Framework
- Layered viewpoints
 - KAOS: requirements
 - mKAOS: missions
- Each diagram (mKAOS model) is organized as a **layer**



The mKAOS Studio tool

toolset for manipulating the model area

palette



Main screen of mKAOS Studio

Example: Flood Monitoring SoS

Global missions

- 1) Detect flood with maximum confidence
- 2) Alert citizens in risky areas



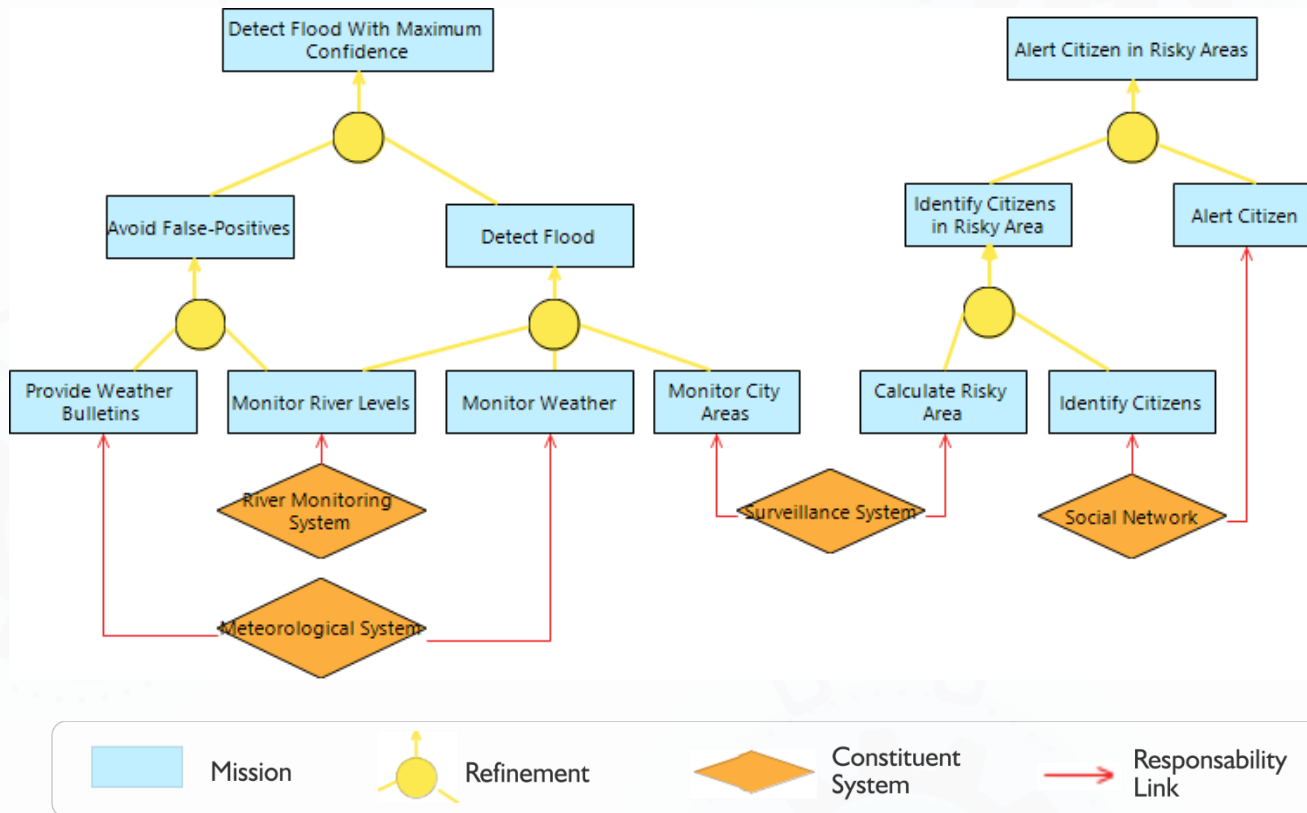
Constituent systems

- 1) River monitoring system
- 2) Meteorological system
- 3) Surveillance system
- 4) Social network



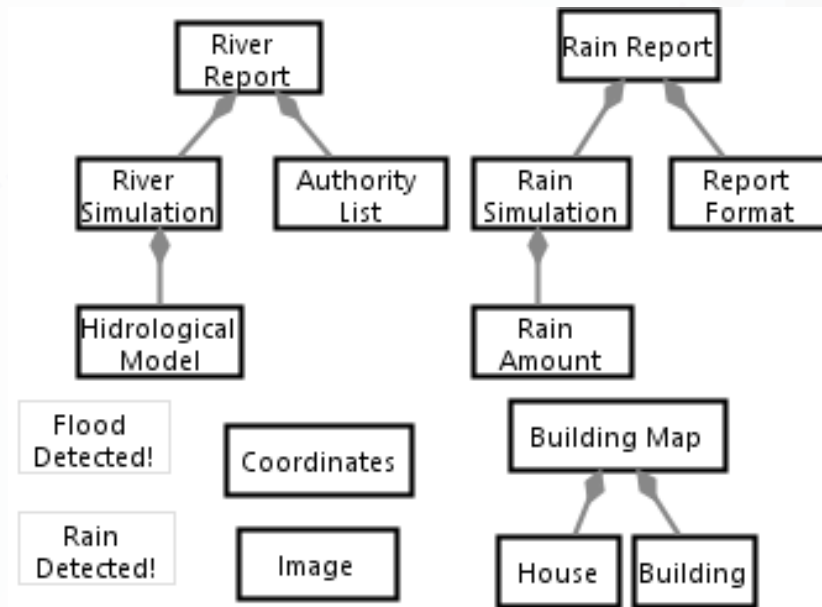
Example: Flood Monitoring SoS

Mission Model + Responsibility Model (overlapped)



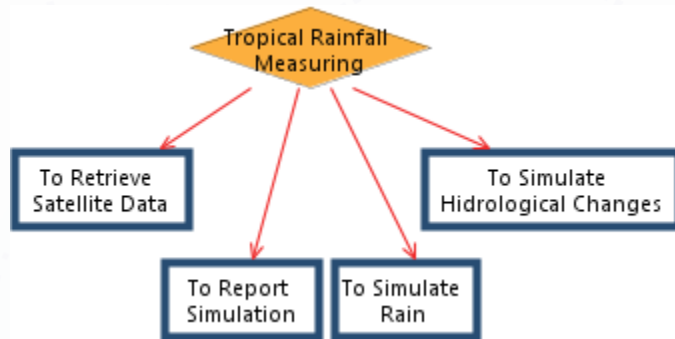
Example: Flood Monitoring SoS

Object Model

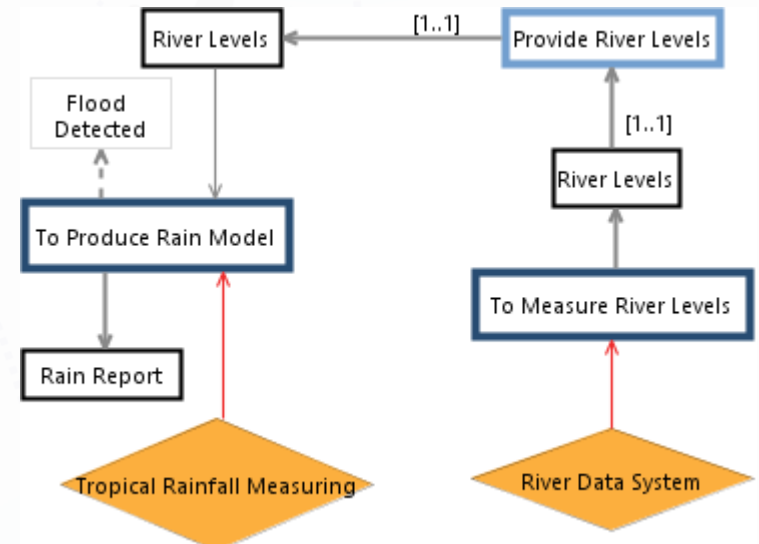


Example: Flood Monitoring SoS

Operational Capability Model

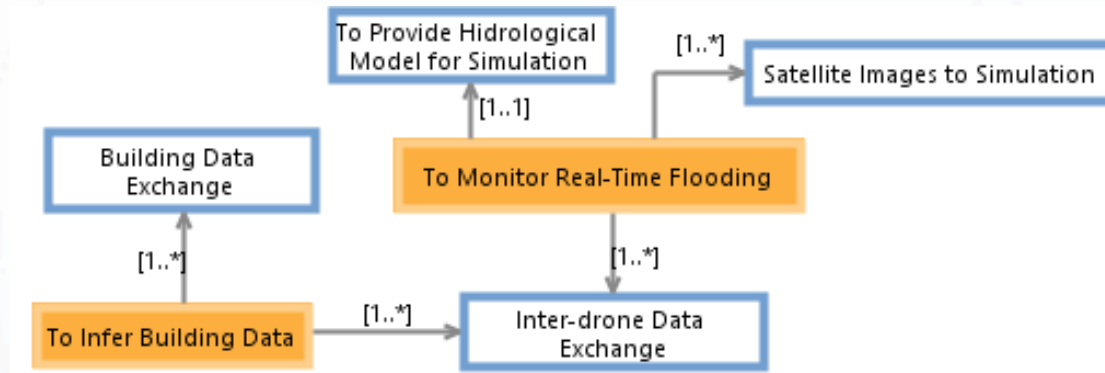


Communicational Capability Model



Example: Flood Monitoring SoS

Emergent Behavior Model



Future work

- Transition from mission to architectural models
 - Integration of the mKAOS language and tool with existing approaches for SoS architectural modeling
- Analysis of mKAOS models
 - *Are all individual missions assigned to the constituent systems?*
 - *Are all constituent systems necessary to accomplish the global missions of the SoS?*

Future work

- **Validation** of the mKAOS language and tool in other SoS scenarios
- **Evaluation of mKAOS with experts**
 - Assessment of the language in terms of **completeness and expressiveness**
 - Assessment of the **applicability of a mission-oriented process** for designing SoS

Any questions?



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